Algebra 4 — Using formulae · Change of subject

Calculator allowed for the numerical parts. Marks in brackets. ★ = harder question.

1 Make x the subject of each equation. [6]

a $d - 4x = f$ **b** $p = 3(q + x)$ **c** $\dfrac{x + a}{b} = c$

a Add $4x$, subtract f: $d - f = 4x \Rightarrow x = \dfrac{d - f}{4}$.

b Divide by 3: $\dfrac{p}{3} = q + x \Rightarrow x = \dfrac{p}{3} - q$ (or $\dfrac{p - 3q}{3}$).

c Multiply by b: $x + a = bc \Rightarrow x = bc - a$.

2 Make x the subject of each formula. [9]

a $x^2 - B = 3C$ **b** $\dfrac{x^2}{y} - z = a$ **c** $p(x - q) = x$

a $x^2 = 3C + B \Rightarrow x = \pm\sqrt{3C + B}$.

b $\dfrac{x^2}{y} = a + z \Rightarrow x^2 = y(a + z) \Rightarrow x = \pm\sqrt{y(a + z)}$.

c x appears twice — expand and collect the x terms on one side:
$px - pq = x \Rightarrow px - x = pq \Rightarrow x(p - 1) = pq \Rightarrow x = \dfrac{pq}{p - 1}$.

3 The formula $s = ut + \frac{1}{2}at^2$ connects distance s, initial speed u, time t and acceleration a. [4]

a Make a the subject.

b Calculate a when $s = 120$ m, $u = 3$ m/s and $t = 6$ s.

a $s - ut = \frac{1}{2}at^2 \Rightarrow a = \dfrac{2(s - ut)}{t^2}$.

b $a = \dfrac{2(120 - 18)}{36} = \dfrac{204}{36} = 5.67$ m/s².

4 A shape is a semicircle of radius r. [6]

a Write a formula for its perimeter P (curved edge plus diameter).

b Write a formula for its area A.

c Find the value of r for which $A = P$ numerically.

a Half a circumference plus the diameter: $P = \pi r + 2r = r(\pi + 2)$.

b $A = \frac{1}{2}\pi r^2$.

c $\frac{1}{2}\pi r^2 = r(\pi + 2)$. Divide by r: $\frac{1}{2}\pi r = \pi + 2 \Rightarrow r = \dfrac{2(\pi + 2)}{\pi} = 3.27$ (3 s.f.).

5 The volume of a cylinder is $V = \pi r^2 h$. A cylinder has volume 500 cm³ and height 10 cm. Find its radius. [3]

> Rearrange: $r^2 = \dfrac{V}{\pi h} \Rightarrow r = \sqrt{\dfrac{V}{\pi h}} = \sqrt{\dfrac{500}{10\pi}} = 3.99$ cm (positive root only — it's a length).

6 Make t the subject of $m = \dfrac{3t+1}{t-2}$. [4]

> Multiply out: $m(t-2) = 3t+1 \Rightarrow mt - 2m = 3t + 1$. Collect t terms: $mt - 3t = 2m + 1 \Rightarrow t(m-3) = 2m+1 \Rightarrow t = \dfrac{2m+1}{m-3}$.

Total: 32 marks

Check: undo operations in reverse order, one step at a time · a square root gives ± unless the quantity must be positive · if the new subject appears twice, collect its terms and factorise it out.